

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457636.7238 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.038
Transit depth (Rp/Rs)^2	1.33 +/- 0.89 [%]
Orbital Inclination (inc)	87.39 +/- 2.1
Airmass coefficient 1 (a1)	0.905 +/- 0.014
Airmass coefficient 2 (a2)	0.095 +/- 0.077
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	#3 - [100, 145]
Optimal Aperture	3.02
Optimal Annulus	11.7
Transit Duration (day)	0.072 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.115 ± 0.038 vs 0.1394 (deviation 0.64σ)
Duration fit vs published	0.072 d vs 0.0983 d (deviation 2.39σ)
Mid-time O–C	5.7 min
Depth SNR	3.0
Residual scatter	1.53 %

Light curve

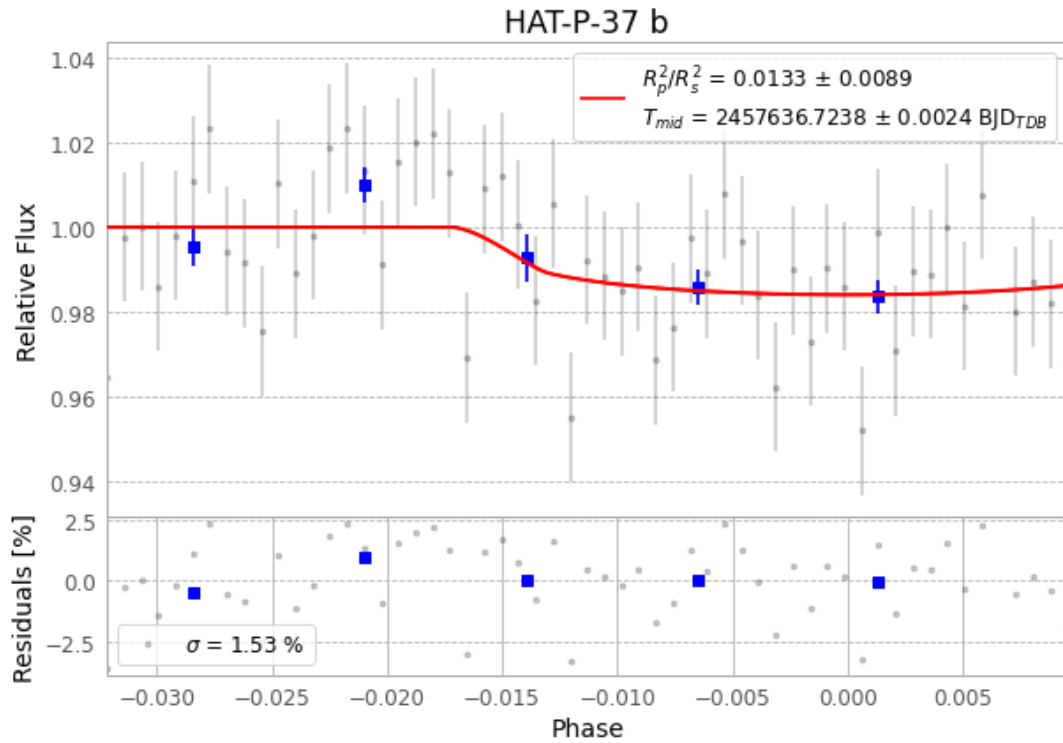


Figure 1 — FinalLightCurve_HAT-P-37 b_2016-09-04.png (SHA-256 12bcb5b3994eaccc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [254, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aea081ba63317188...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), HATP-37160905030912.FITS → HATP-37160905055712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-160905.log (c8a9dc368a00...), FinalLightCurve_HAT-P-37 b_2016-09-04.png (12bcb5b3994e...), FinalLightCurve_HAT-P-37 b_2016-09-04.csv (2578babefb12...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:55Z.